

Regolith Evolved Gas Analyzer (REGA)



**An Instrument to Study
Martian Soil Mineralogy and Atmospheric Composition**

**Presented at
Third Harsh Environment Mass Spectrometry Conference
Pasadena, CA**

**by
John H. Hoffman
Physics Department
University of Texas at Dallas**

March 26, 2002

Regolith Evolved Gas Analyzer (REGA)



What is REGA?

- **REGA is an instrument designed for operation on the surface of Mars to:**
 - **Measure and determine the mineralogical composition of soil samples.**
 - **Measure and determine the reactivity of soils.**
- **REGA identifies and quantifies by mass spectroscopic analysis the gases and vapors released from programmed heating of soils.**
- **REGA provides long term monitoring of atmospheric composition.**

Regolith Evolved Gas Analyzer (REGA)

REGA Science

- Expected results from operation on surface of Mars:
 - **Mineralogical composition of soils.**
 - **Chemical reactivity of soils.**
 - **Composition of Atmosphere.**
- Lead to:
 - **Understanding of geological and climatological histories of Mars.**
 - **Identifying prerequisite components for existence of life.**
 - **Specifically finding existence of carbonates and evaporites- these contain water, the prerequisite for life.**
 - **Assessing possible environmental hazards to be encountered by future human explorers and their equipment.**

Regolith Evolved Gas Analyzer (REGA)



REGA Instrument Operation

An external scoop or drill deposits soil samples into the instrument sample hopper.

Sample hopper contains sieve that separates particles of < 500 μm size and meters them into a fresh stainless steel crucible.

Sample size 0.1 to 0.3 cc.

Mechanical arm moves crucible to furnace, inserts it in furnace, closes furnace.

Sample is heated to 900 °C in stepwise manner.

Gases evolved from soil are transferred to mass spectrometer for analysis.

Regolith Evolved Gas Analyzer (REGA)

REGA Soil Handling System

■ Description

Consists of sample hopper with soil sifter and volume metering mechanism.

Storage rack for 30 stainless steel crucibles .

Three stepper motors to:

- drive mechanical arm to move crucibles from soil handling system to furnace and used crucible from furnace.**
- open and close furnace.**
- operate soil sifter and metering.**
- dump excess bulk soil in hopper out side chute.**

Regolith Evolved Gas Analyzer (REGA)

REGA High Temperature Furnace

- | Cylindrical furnace heats soil sample and crucible from ambient temperature to 900°C following programmable temperature profile.**
- | Efficient insulation minimizes heater power - only 26 watts required to reach 900 °C.**
- | Metal-to-metal seal between upper and lower furnace sections seals in evolved gases.**
- | Heated transfer tube moves controlled amounts of gasses to mass spectrometer for analysis.**
- | Valved exhaust tube vents evolved gases to ambient atmosphere to control pressure in furnace.**

Regolith Evolved Gas Analyzer (REGA)

REGA Mass Spectrometer

■ Description

- Double-focusing magnetic sector-field analyzer**
- Ceramic electron multiplier detectors**
- Weight of analyzer- 900 gms**
- Performance parameters**
 - | Mass range: 2 - 200 2 channel**
 - | Mass resolution: $m/\Delta m = 200$**
 - | Dynamic range: 5×10^6**
 - | Sensitivity: 5 ppb (using isotopic peaks)**
 - | Spectral scan: Adjustable with microprocessor**
 - | Output signal: Digital - Counting rate for each mass peak in spectrum**

Regolith Evolved Gas Analyzer (REGA)

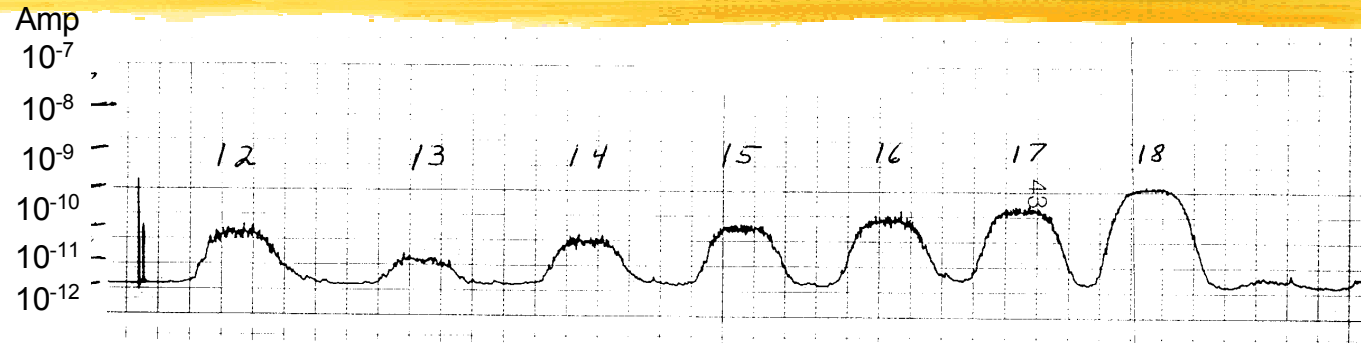
REGA Mass Spectrometer

■ Operation

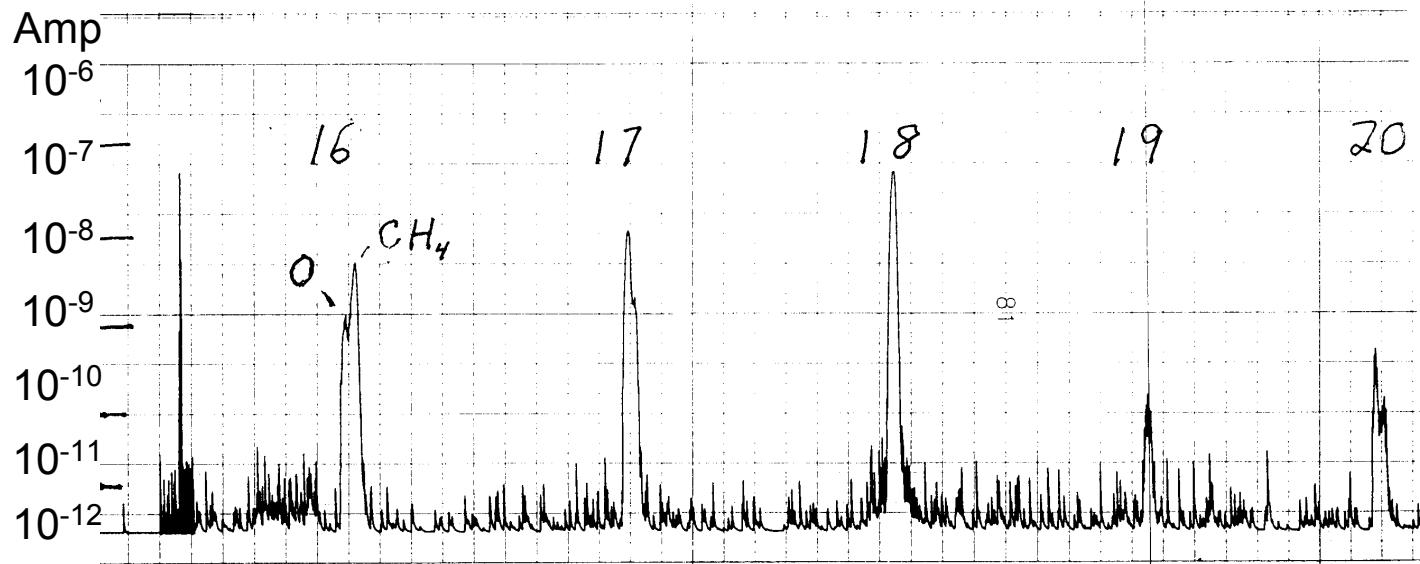
- Gases evolved from heated samples in furnace transferred to mass spectrometer ion source through a heated tube containing a constriction acting as a leak.**
- Ion source region separated from mass analyzer via object slit of ion optics.**
- Pressure in source region maintained at 1×10^{-5} torr.**
- Analyzer section pumped by getter and miniature ion pump. Pressure at 10^{-8} torr**
- Ion source:**
 - 4 electron beam energies from 90 to 20 eV.**
 - 2 emission currents: 250 and 25 microamps.**

Regolith Evolved Gas Analyzer (REGA)

Low mass



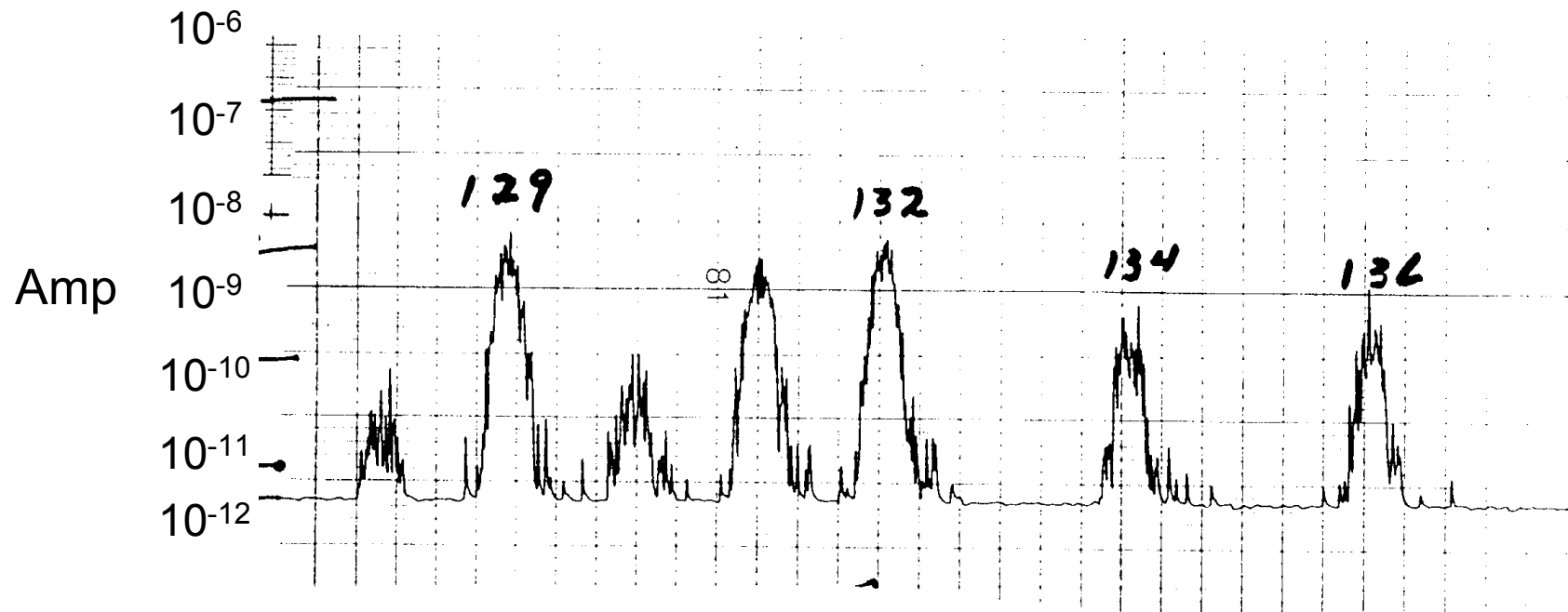
High mass



Mass spectra on low and high mass channels

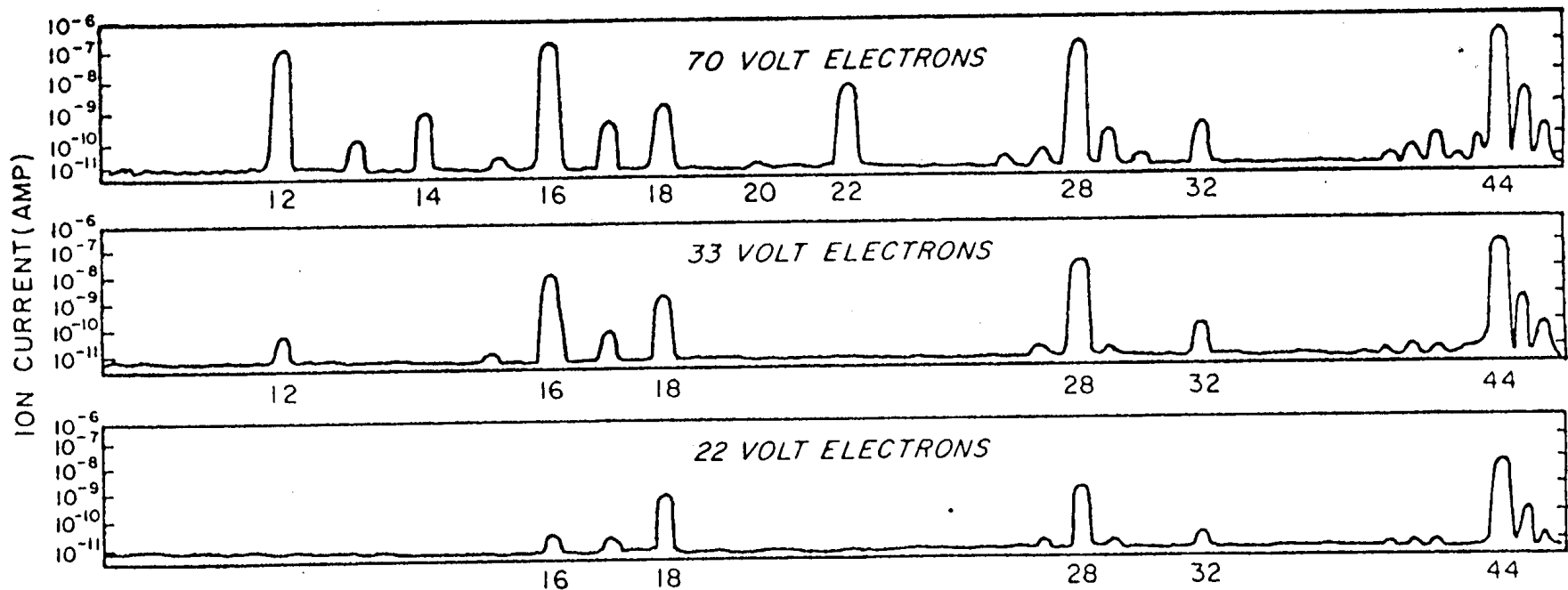
J. Hoffman, University of Texas at Dallas

Regolith Evolved Gas Analyzer (REGA)



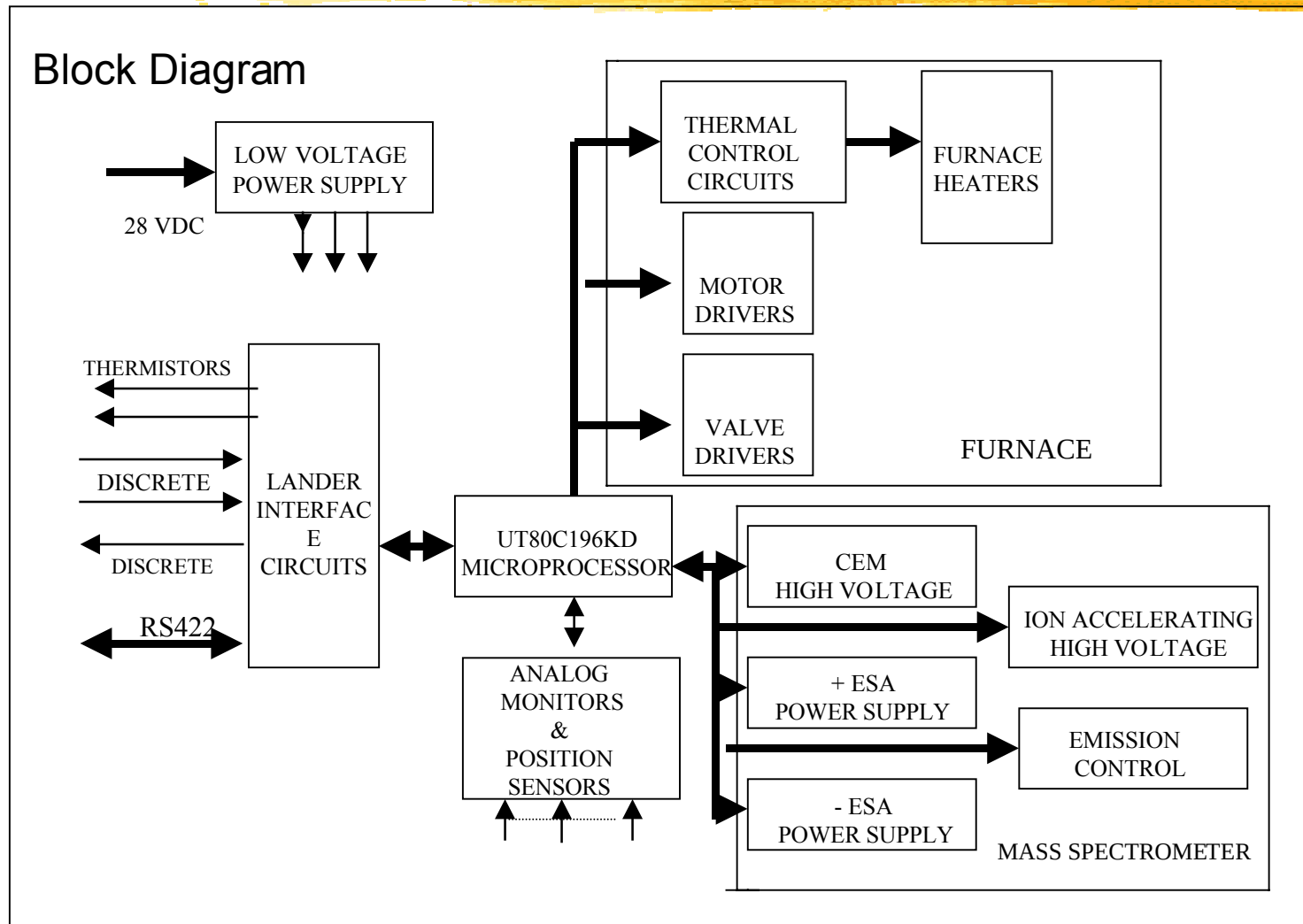
Xenon mass spectrum

Regolith Evolved Gas Analyzer (REGA)



CO₂ spectra at several electron energy levels

Regolith Evolved Gas Analyzer (REGA)



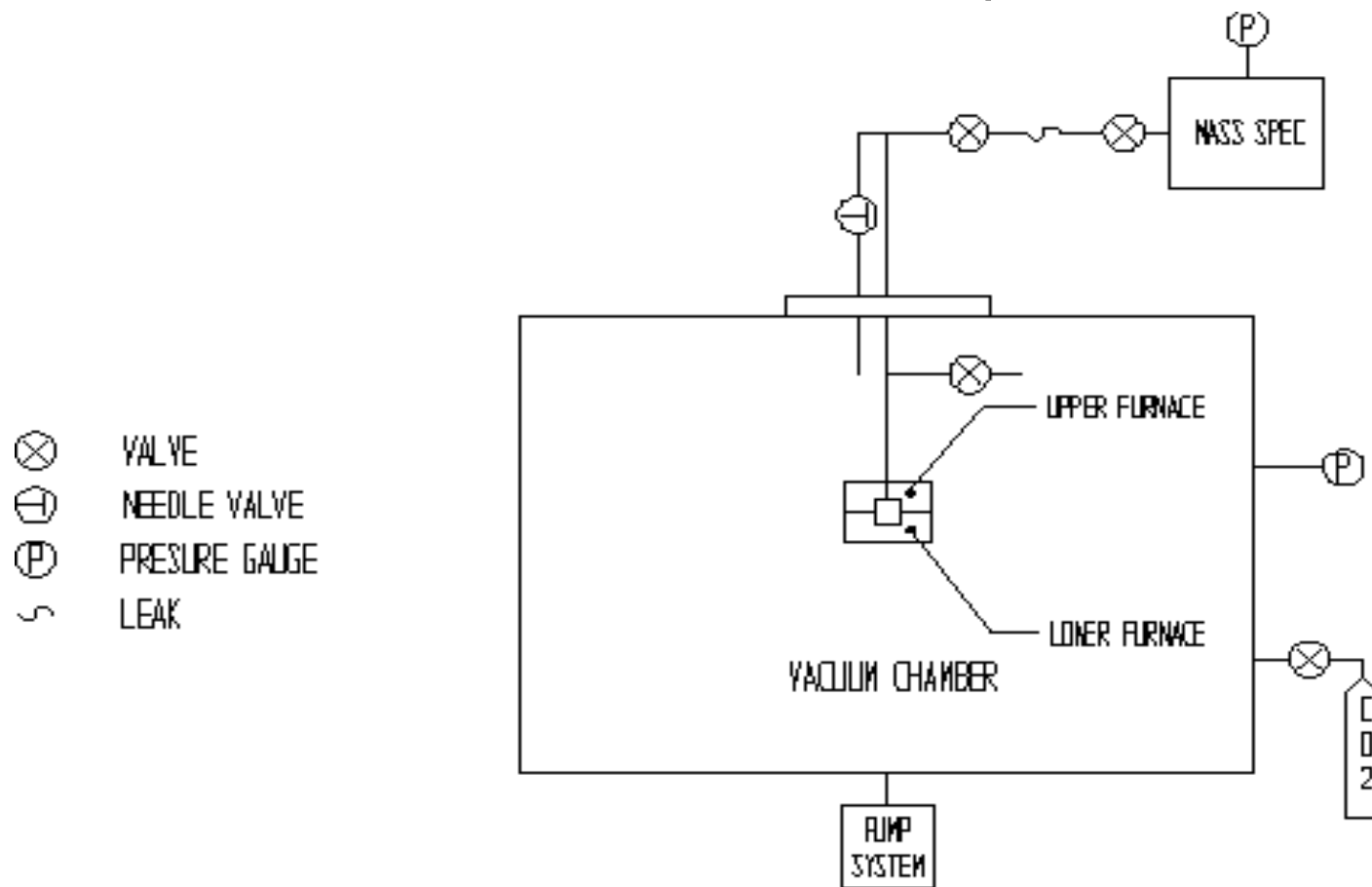
Regolith Evolved Gas Analyzer (REGA)



J. Hoffman, University of Texas at Dallas

Regolith Evolved Gas Analyzer (REGA)

REGA Test Set-up

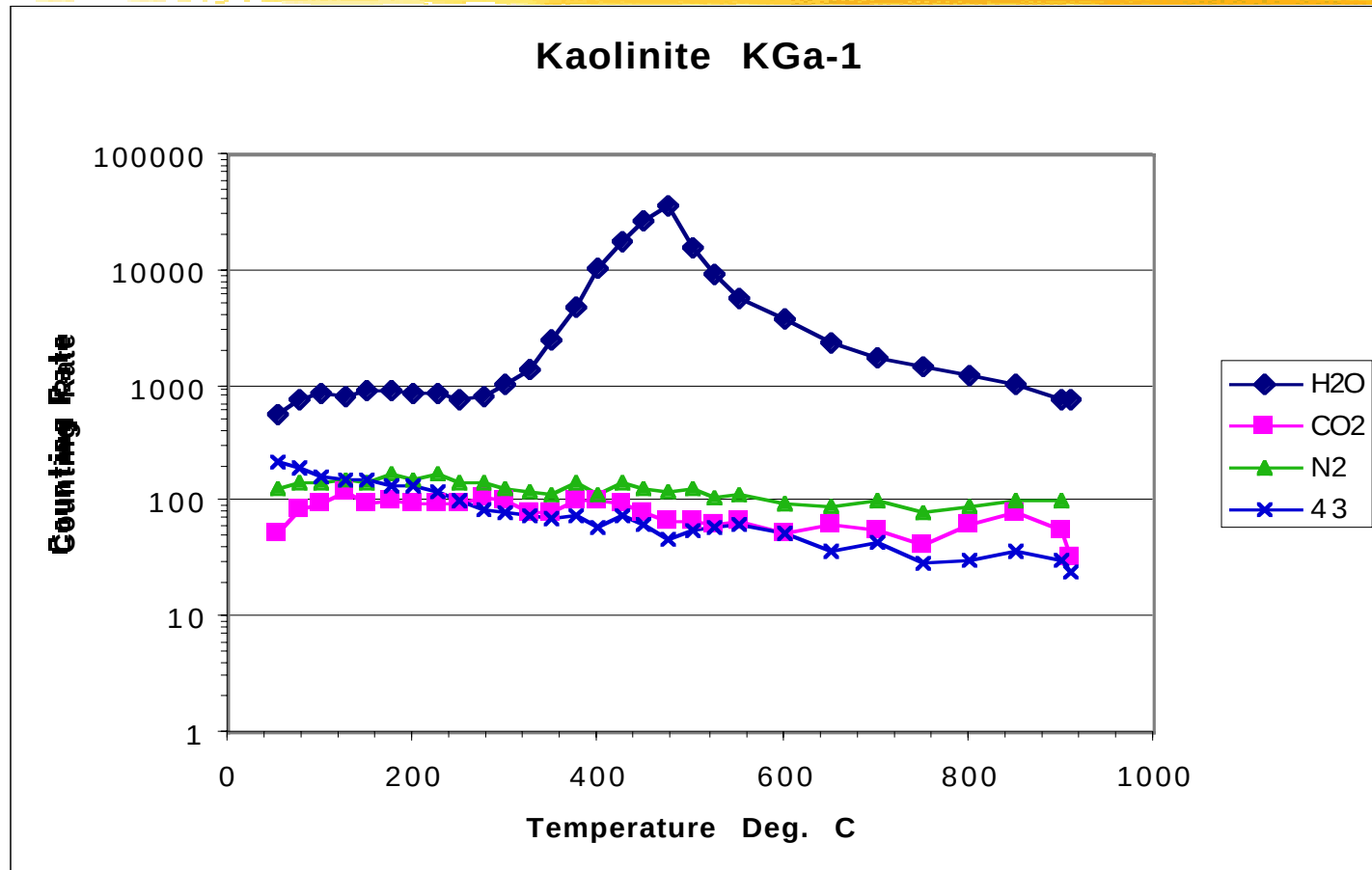


Regolith Evolved Gas Analyzer (REGA)

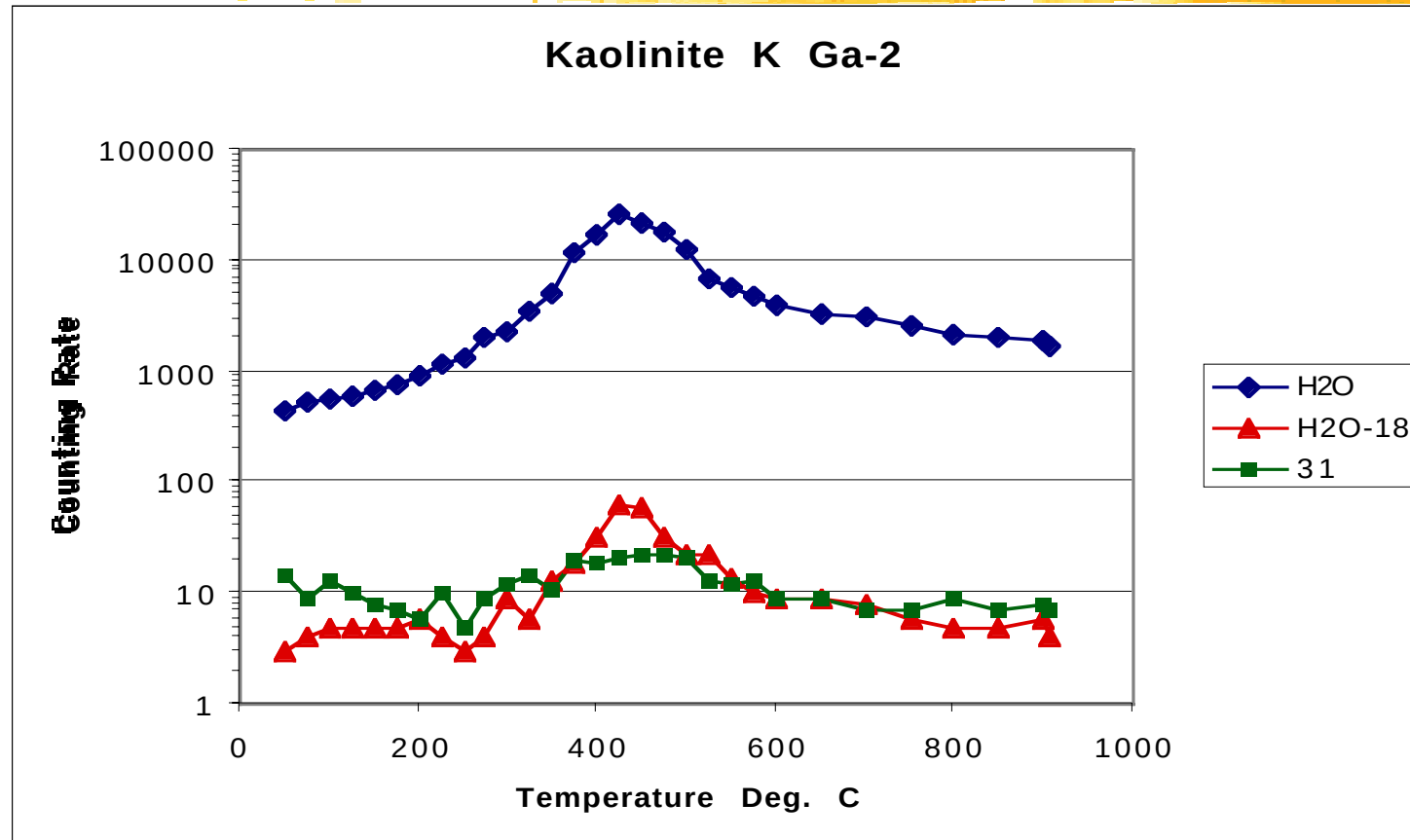


J. Hoffman, University of Texas at Dallas

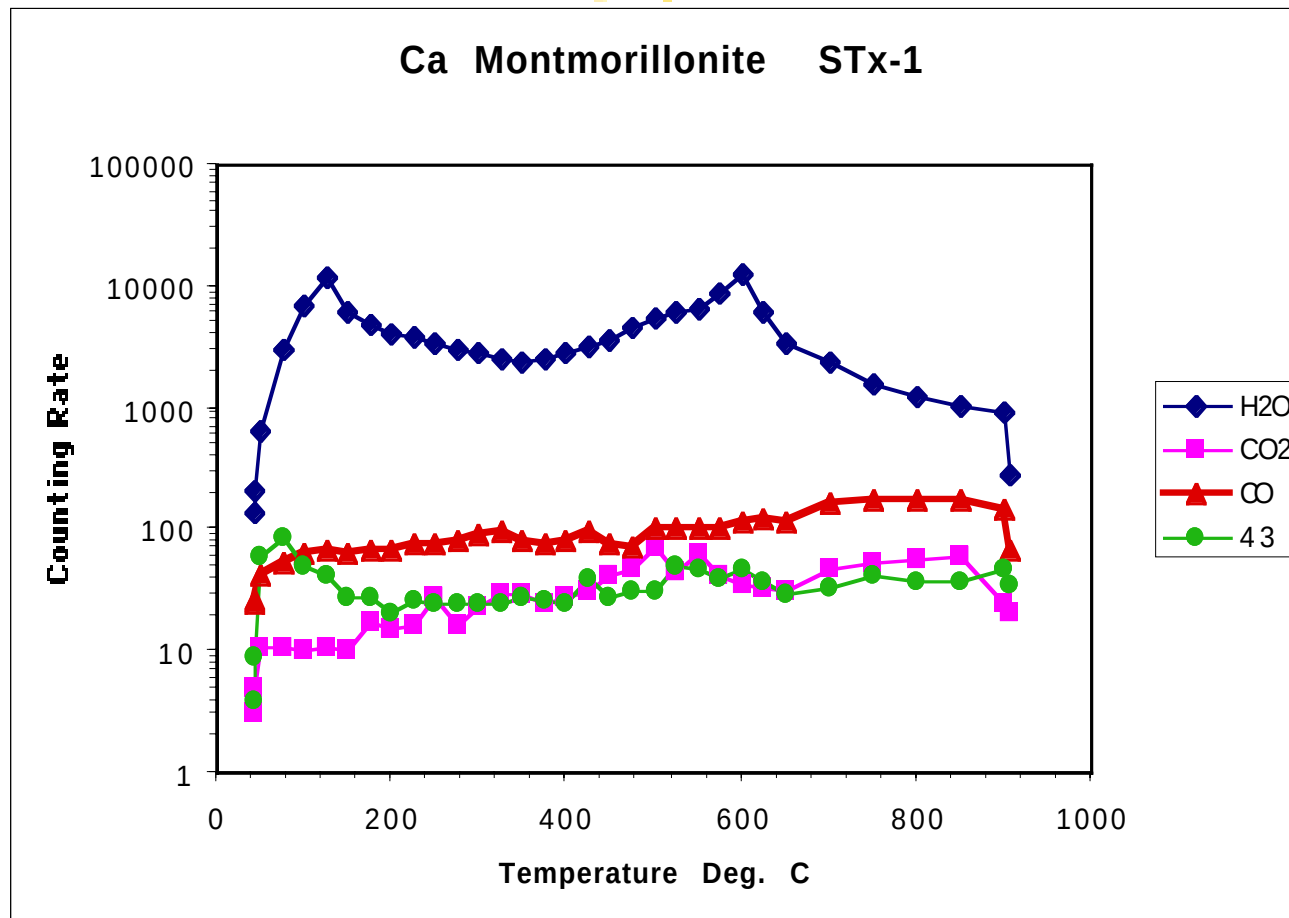
Regolith Evolved Gas Analyzer (REGA)



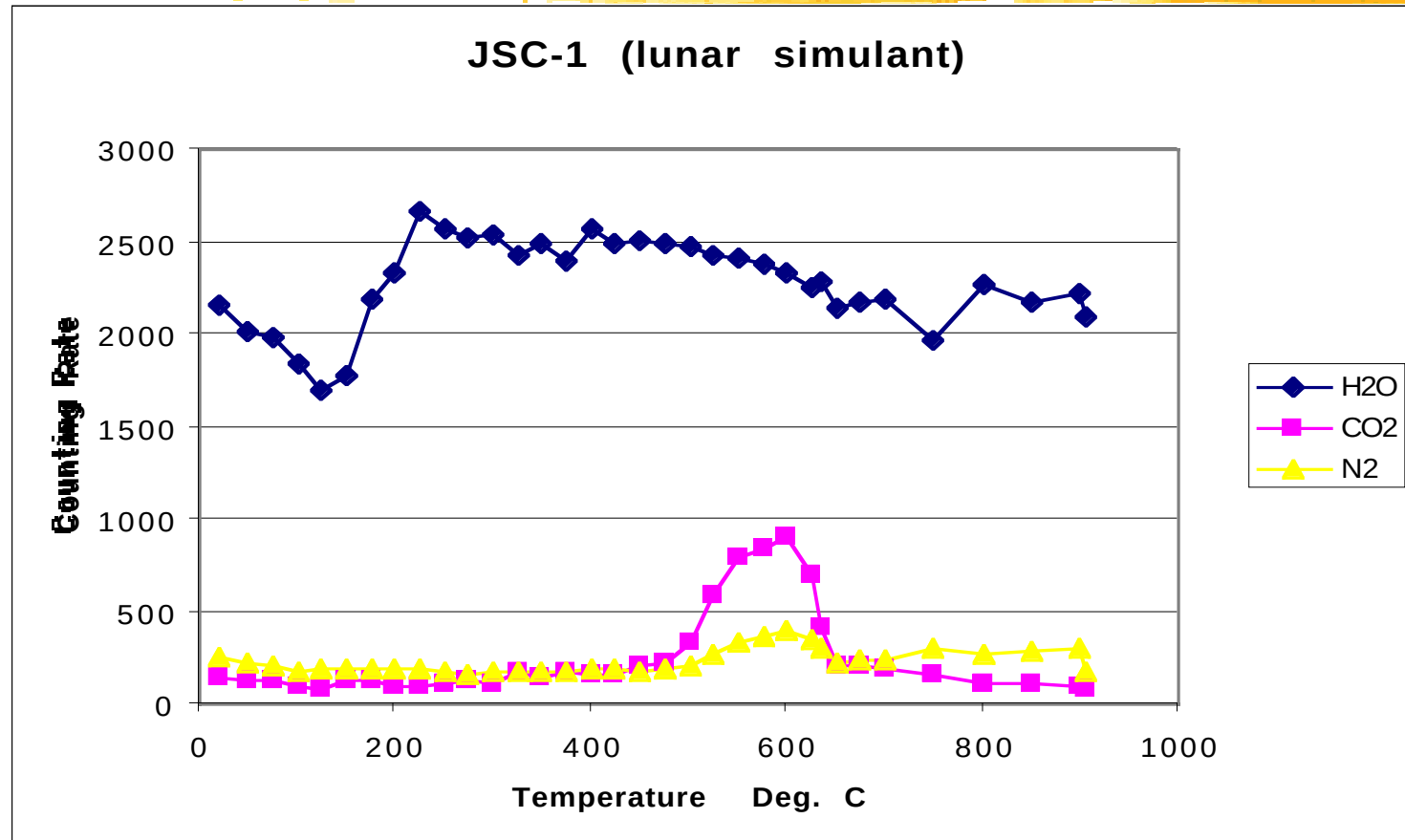
Regolith Evolved Gas Analyzer (REGA)



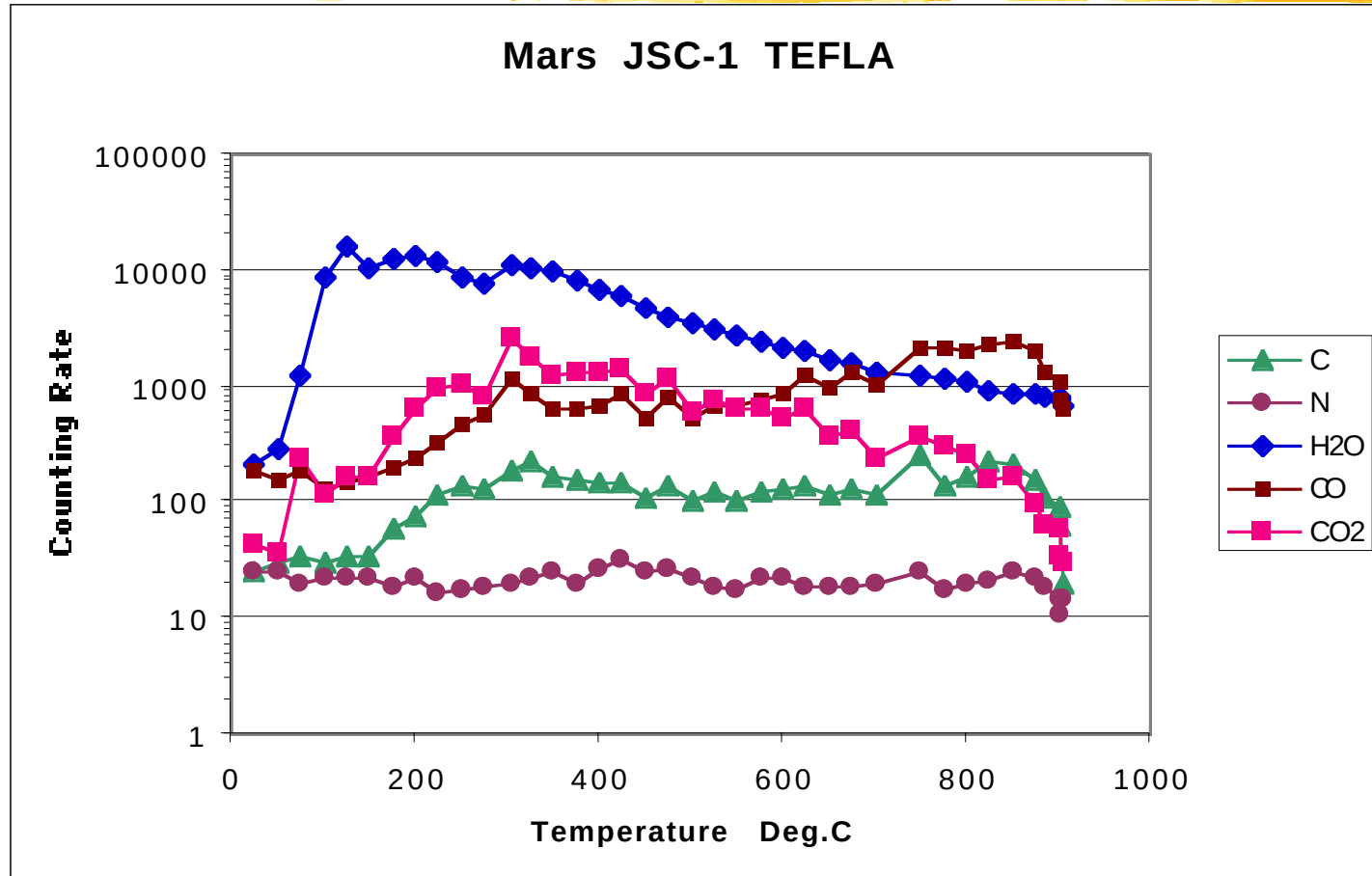
Regolith Evolved Gas Analyzer (REGA)



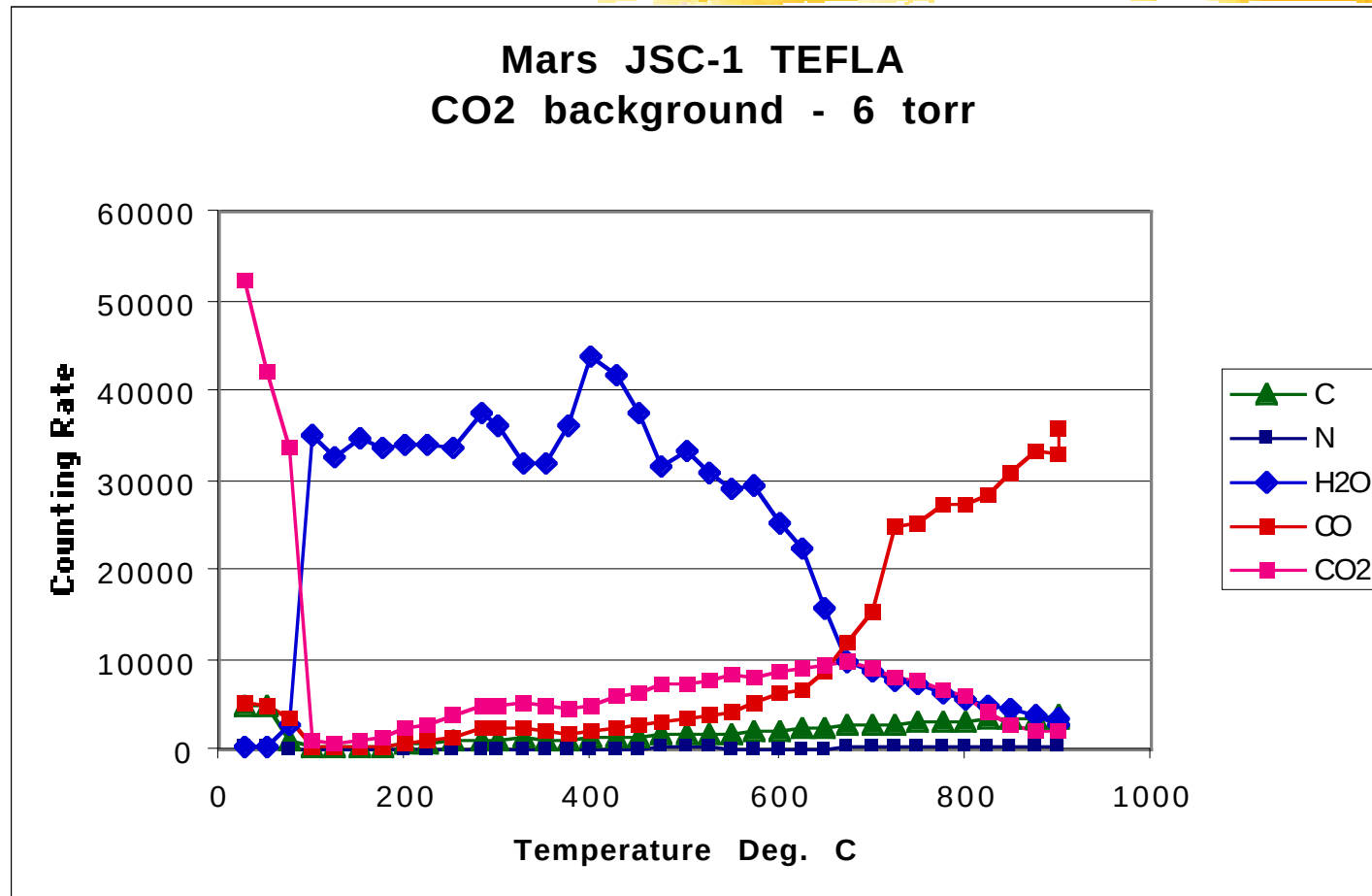
Regolith Evolved Gas Analyzer (REGA)



Regolith Evolved Gas Analyzer (REGA)



Regolith Evolved Gas Analyzer (REGA)



Regolith Evolved Gas Analyzer (REGA)



Conclusion

- Demonstration:
 - REGA instrument's capability to extract and measure volatile materials from simulated martian soils.
- Development stage:
 - Software for operation of furnace system and mass spectrometer.
 - Instrument model for further tests with soil simulants.